

- (a) Show that $(fg)^*(\alpha) = f^*(\alpha) + g^*(\alpha)$ for primitive elements $\alpha \in H^*(X; R)$.
- (b) Deduce that the k^{th} -power map $x \mapsto x^k$ induces the map $\alpha \mapsto k\alpha$ on primitive elements α . In particular the quaternionic k^{th} -power map $S^3 \rightarrow S^3$ has degree k .
- (c) Show that every polynomial $a_n x^n b_n + \cdots + a_1 x b_1 + a_0$ of nonzero degree with coefficients in $\mathbb{H}$ has a root in $\mathbb{H}$. [See Theorem 1.8.]
11. If T^n is the n -dimensional torus, the product of n circles, show that the Pontryagin ring $H_*(T^n; \mathbb{Z})$ is the exterior algebra $\Lambda_{\mathbb{Z}}[x_1, \dots, x_n]$ with $|x_i| = 1$.
12. Compute the Pontryagin product structure in $H_*(L; \mathbb{Z}_p)$ where L is an infinite-dimensional lens space $S^\infty/\mathbb{Z}_p$, for p an odd prime, using the coproduct in $H^*(L; \mathbb{Z}_p)$.
13. Verify that the Hopf algebras $\Lambda_R[\alpha]$ and $\mathbb{Z}_p[\alpha]/(\alpha^p)$ are self-dual.
14. Show that the coproduct in the Hopf algebra $H_*(X; R)$ dual to $H^*(X; R)$ is induced by the diagonal map $X \rightarrow X \times X$, $x \mapsto (x, x)$.
15. Suppose that X is a path-connected H-space such that $H^*(X; \mathbb{Z})$ is free and finitely generated in each dimension, and $H^*(X; \mathbb{Q})$ is a polynomial ring $\mathbb{Q}[\alpha]$. Show that the Pontryagin ring $H_*(X; \mathbb{Z})$ is commutative and associative, with a structure uniquely determined by the ring $H^*(X; \mathbb{Z})$.
16. Classify algebraically the Hopf algebras A over $\mathbb{Z}$ such that A^n is free for each n and $A \otimes \mathbb{Q} \approx \mathbb{Q}[\alpha]$. In particular, determine which Hopf algebras $A \otimes \mathbb{Z}_p$ arise from such A 's.

3.D The Cohomology of $SO(n)$

After the general discussion of homological and cohomological properties of H-spaces in the preceding section, we turn now to a family of quite interesting and subtle examples, the orthogonal groups $O(n)$. We will compute their homology and cohomology by constructing very nice CW structures on them, and the results illustrate the general structure theorems of the last section quite well. After dealing with the orthogonal groups we then describe the straightforward generalization to Stiefel manifolds, which are also fairly basic objects in algebraic and geometric topology.

The orthogonal group $O(n)$ can be defined as the group of isometries of $\mathbb{R}^n$ fixing the origin. Equivalently, this is the group of $n \times n$ matrices A with entries in $\mathbb{R}$ such that $AA^t = I$, where A^t is the transpose of A . From this viewpoint, $O(n)$ is topologized as a subspace of $\mathbb{R}^{n^2}$, with coordinates the n^2 entries of an $n \times n$ matrix. Since the columns of a matrix in $O(n)$ are unit vectors, $O(n)$ can also be regarded as a subspace of the product of n copies of S^{n-1} . It is a closed subspace since the conditions that columns be orthogonal are defined by polynomial equations. Hence

$O(n)$ is compact. The map $O(n) \times O(n) \rightarrow O(n)$ given by matrix multiplication is continuous since it is defined by polynomials. The inversion map $A \mapsto A^{-1} = A^t$ is clearly continuous, so $O(n)$ is a topological group, and in particular an H-space.

The determinant map $O(n) \rightarrow \{\pm 1\}$ is a surjective homomorphism, so its kernel $SO(n)$, the ‘special orthogonal group,’ is a subgroup of index two. The two cosets $SO(n)$ and $O(n) - SO(n)$ are homeomorphic to each other since for fixed $B \in O(n)$ of determinant -1 , the maps $A \mapsto AB$ and $A \mapsto AB^{-1}$ are inverse homeomorphisms between these two cosets. The subgroup $SO(n)$ is a union of components of $O(n)$ since the image of the map $O(n) \rightarrow \{\pm 1\}$ is discrete. In fact, $SO(n)$ is path-connected since by linear algebra, each $A \in SO(n)$ is a rotation, a composition of rotations in a family of orthogonal 2-dimensional subspaces of $\mathbb{R}^n$, with the identity map on the subspace orthogonal to all these planes, and such a rotation can obviously be joined to the identity by a path of rotations of the same planes through decreasing angles. Another reason why $SO(n)$ is connected is that it has a CW structure with a single 0-cell, as we show in Proposition 3D.1. An exercise at the end of the section is to show that a topological group with a CW structure is an orientable manifold, so $SO(n)$ is a closed orientable manifold. From the CW structure it follows that its dimension is $n(n-1)/2$. These facts can also be proved using fiber bundles.

The group $O(n)$ is a subgroup of $GL_n(\mathbb{R})$, the ‘general linear group’ of all invertible $n \times n$ matrices with entries in $\mathbb{R}$, discussed near the beginning of §3.C. The Gram-Schmidt orthogonalization process applied to the columns of matrices in $GL_n(\mathbb{R})$ provides a retraction $r: GL_n(\mathbb{R}) \rightarrow O(n)$, continuity of r being evident from the explicit formulas for the Gram-Schmidt process. By inserting appropriate scalar factors into these formulas it is easy to see that $O(n)$ is in fact a deformation retract of $GL_n(\mathbb{R})$. Using a bit more linear algebra, namely the polar decomposition, it is possible to show that $GL_n(\mathbb{R})$ is actually homeomorphic to $O(n) \times \mathbb{R}^k$ for $k = n(n+1)/2$.

The topological structure of $SO(n)$ for small values of n can be described in terms of more familiar spaces:

- $SO(1)$ is a point.
- $SO(2)$, the rotations of $\mathbb{R}^2$, is both homeomorphic and isomorphic as a group to S^1 , thought of as the unit complex numbers.
- $SO(3)$ is homeomorphic to $\mathbb{RP}^3$. To see this, let $\varphi: D^3 \rightarrow SO(3)$ send a nonzero vector x to the rotation through angle $|x|\pi$ about the axis formed by the line through the origin in the direction of x . An orientation convention such as the ‘right-hand rule’ is needed to make this unambiguous. By continuity, φ then sends 0 to the identity. Antipodal points of $S^2 = \partial D^3$ are sent to the same rotation through angle π , so φ induces a map $\overline{\varphi}: \mathbb{RP}^3 \rightarrow SO(3)$, regarding $\mathbb{RP}^3$ as D^3 with antipodal boundary points identified. The map $\overline{\varphi}$ is clearly injective since the axis of a nontrivial rotation is uniquely determined as its fixed point set, and $\overline{\varphi}$ is surjective since by easy linear algebra each nonidentity element

of $SO(3)$ is a rotation about some axis. It follows that $\overline{\varphi}$ is a homeomorphism $\mathbb{RP}^3 \approx SO(3)$.

- $SO(4)$ is homeomorphic to $S^3 \times SO(3)$. Identifying $\mathbb{R}^4$ with the quaternions $\mathbb{H}$ and S^3 with the group of unit quaternions, the quaternion multiplication $v \mapsto vw$ for fixed $w \in S^3$ defines an isometry $\rho_w \in O(4)$ since $|vw| = |v||w| = |v|$ if $|w| = 1$. Points of $O(4)$ are 4-tuples $(v_1, \dots, v_4)$ of orthonormal vectors $v_i \in \mathbb{H} = \mathbb{R}^4$, and we view $O(3)$ as the subspace with $v_1 = 1$. A homeomorphism $S^3 \times O(3) \rightarrow O(4)$ is defined by sending $(v, (1, v_2, v_3, v_4))$ to $(v, v_2v, v_3v, v_4v) = \rho_v(1, v_2, v_3, v_4)$, with inverse $(v, v_2, v_3, v_4) \mapsto (v, (1, v_2v^{-1}, v_3v^{-1}, v_4v^{-1})) = (v, \rho_{v^{-1}}(v, v_2, v_3, v_4))$. Restricting to identity components, we obtain a homeomorphism $S^3 \times SO(3) \approx SO(4)$. This is not a group isomorphism, however. It can be shown, though we will not digress to do so here, that the homomorphism $\psi: S^3 \times S^3 \rightarrow SO(4)$ sending a pair (u, v) of unit quaternions to the isometry $w \mapsto uwv^{-1}$ of $\mathbb{H}$ is surjective with kernel $\mathbb{Z}_2 = \{\pm(1, 1)\}$, and that ψ is a covering space projection, representing $S^3 \times S^3$ as a 2-sheeted cover of $SO(4)$, the universal cover. Restricting ψ to the diagonal $S^3 = \{(u, u)\} \subset S^3 \times S^3$ gives the universal cover $S^3 \rightarrow SO(3)$, so $SO(3)$ is isomorphic to the quotient group of S^3 by the normal subgroup $\{\pm 1\}$.

Using octonions one can construct in the same way a homeomorphism $SO(8) \approx S^7 \times SO(7)$. But in all other cases $SO(n)$ is only a ‘twisted product’ of $SO(n-1)$ and S^{n-1} ; see Example 4.55 and the discussion following Corollary 4D.3.

Cell Structure

Our first task is to construct a CW structure on $SO(n)$. This will come with a very nice cellular map $\rho: \mathbb{RP}^{n-1} \times \mathbb{RP}^{n-2} \times \dots \times \mathbb{RP}^1 \rightarrow SO(n)$. To simplify notation we will write P^i for $\mathbb{RP}^i$.

To each nonzero vector $v \in \mathbb{R}^n$ we can associate the reflection $r(v) \in O(n)$ across the hyperplane consisting of all vectors orthogonal to v . Since $r(v)$ is a reflection, it has determinant -1 , so to get an element of $SO(n)$ we consider the composition $\rho(v) = r(v)r(e_1)$ where e_1 is the first standard basis vector $(1, 0, \dots, 0)$. Since $\rho(v)$ depends only on the line spanned by v , ρ defines a map $P^{n-1} \rightarrow SO(n)$. This map is injective since it is the composition of $v \mapsto r(v)$, which is obviously an injection of P^{n-1} into $O(n) - SO(n)$, with the homeomorphism $O(n) - SO(n) \rightarrow SO(n)$ given by right-multiplication by $r(e_1)$. Since ρ is injective and P^{n-1} is compact Hausdorff, we may think of ρ as embedding P^{n-1} as a subspace of $SO(n)$.

More generally, for a sequence $I = (i_1, \dots, i_m)$ with each $i_j < n$, we define a map $\rho: P^I = P^{i_1} \times \dots \times P^{i_m} \rightarrow SO(n)$ by letting $\rho(v_1, \dots, v_m)$ be the composition $\rho(v_1) \dots \rho(v_m)$. If $\varphi^i: D^i \rightarrow P^i$ is the standard characteristic map for the i -cell of P^i , restricting to the 2-sheeted covering projection $\partial D^i \rightarrow P^{i-1}$, then the product $\varphi^I: D^I \rightarrow P^I$ of the appropriate φ^{i_j} ’s is a characteristic map for the top-dimensional

cell of P^I . We will be especially interested in the sequences $I = (i_1, \dots, i_m)$ satisfying $n > i_1 > \dots > i_m > 0$. These sequences will be called *admissible*, as will the sequence consisting of a single 0.

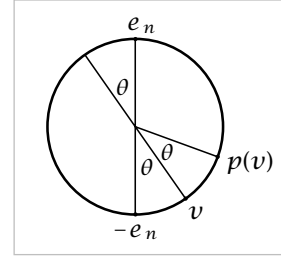
Proposition 3D.1. *The maps $\rho\varphi^I: D^I \rightarrow SO(n)$, for I ranging over all admissible sequences, are the characteristic maps of a CW structure on $SO(n)$ for which the map $\rho: P^{n-1} \times P^{n-2} \times \dots \times P^1 \rightarrow SO(n)$ is cellular.*

In particular, there is a single 0-cell $e^0 = \{\mathbb{1}\}$, so $SO(n)$ is path-connected. The other cells $e^I = e^{i_1} \dots e^{i_m}$ are products, via the group operation in $SO(n)$, of the cells $e^i \subset P^{n-1} \subset SO(n)$.

Proof: According to Proposition A.2 in the Appendix, there are three things to show in order to obtain the CW structure:

- (1) For each decreasing sequence I , $\rho\varphi^I$ is a homeomorphism from the interior of D^I onto its image.
- (2) The resulting image cells e^I are all disjoint and cover $SO(n)$.
- (3) For each e^I , $\rho\varphi^I(\partial D^I)$ is contained in a union of cells of lower dimension than e^I .

To begin the verification of these properties, define $p: SO(n) \rightarrow S^{n-1}$ by evaluation at the vector $e_n = (0, \dots, 0, 1)$, $p(\alpha) = \alpha(e_n)$. Isometries in $P^{n-2} \subset P^{n-1} \subset SO(n)$ fix e_n , so $p(P^{n-2}) = \{e_n\}$. We claim that p is a homeomorphism from $P^{n-1} - P^{n-2}$ onto $S^{n-1} - \{e_n\}$. This can be seen as follows. Thinking of a point in P^{n-1} as a vector v , the map p takes this to $\rho(v)(e_n) = r(v)r(e_1)(e_n)$, which equals $r(v)(e_n)$ since e_n is in the hyperplane orthogonal to e_1 . From the picture at the right it is then clear that p simply stretches the lower half of each meridian circle in S^{n-1} onto the whole meridian circle, doubling the angle up from the south pole, so $P^{n-1} - P^{n-2}$, represented by vectors whose last coordinate is negative, is taken homeomorphically onto $S^{n-1} - \{e_n\}$.



The next statement is that the map

$$h: (P^{n-1} \times SO(n-1), P^{n-2} \times SO(n-1)) \rightarrow (SO(n), SO(n-1)), \quad h(v, \alpha) = \rho(v)\alpha$$

is a homeomorphism from $(P^{n-1} - P^{n-2}) \times SO(n-1)$ onto $SO(n) - SO(n-1)$. Here we view $SO(n-1)$ as the subgroup of $SO(n)$ fixing the vector e_n . To construct an inverse to this homeomorphism, let $\beta \in SO(n) - SO(n-1)$ be given. Then $\beta(e_n) \neq e_n$ so by the preceding paragraph there is a unique $v_\beta \in P^{n-1} - P^{n-2}$ with $\rho(v_\beta)(e_n) = \beta(e_n)$, and v_β depends continuously on β since $\beta(e_n)$ does. The composition $\alpha_\beta = \rho(v_\beta)^{-1}\beta$ then fixes e_n , hence lies in $SO(n-1)$. Since $\rho(v_\beta)\alpha_\beta = \beta$, the map $\beta \mapsto (v_\beta, \alpha_\beta)$ is an inverse to h on $SO(n) - SO(n-1)$.

Statements (1) and (2) can now be proved by induction on n . The map ρ takes P^{n-2} to $SO(n-1)$, so we may assume inductively that the maps $\rho\varphi^I$ for I ranging

over admissible sequences with first term $i_1 < n - 1$ are the characteristic maps for a CW structure on $SO(n - 1)$, with cells the corresponding products e^I . The admissible sequences I with $i_1 = n - 1$ then give disjoint cells e^I covering $SO(n) - SO(n - 1)$ by what was shown in the previous paragraph. So (1) and (2) hold for $SO(n)$.

To prove (3) it suffices to show there is an inclusion $P^i P^i \subset P^i P^{i-1}$ in $SO(n)$ since for an admissible sequence I , the map $\rho: P^I \rightarrow SO(n)$ takes the boundary of the top-dimensional cell of P^I to the image of products P^J with J obtained from I by decreasing one term i_j by 1, yielding a sequence which is admissible except perhaps for having two successive terms equal. As a preliminary to showing that $P^i P^i \subset P^i P^{i-1}$, observe that for $\alpha \in O(n)$ we have $r(\alpha(v)) = \alpha r(v) \alpha^{-1}$. Hence $\rho(v) \rho(w) = r(v) r(e_1) r(w) r(e_1) = r(v) r(w')$ where $w' = r(e_1) w$. Thus to show $P^i P^i \subset P^i P^{i-1}$ it suffices to find for each pair $v, w \in \mathbb{R}^{i+1}$ a pair $x \in \mathbb{R}^{i+1}$, $y \in \mathbb{R}^i$ with $r(v) r(w) = r(x) r(y)$.

Let $V \subset \mathbb{R}^{i+1}$ be a 2-dimensional subspace containing v and w . Since $V \cap \mathbb{R}^i$ is at least 1-dimensional, we can choose a unit vector $y \in V \cap \mathbb{R}^i$. Let $\alpha \in O(i+1)$ take V to $\mathbb{R}^2$ and y to e_1 . Then the conjugate $\alpha r(v) r(w) \alpha^{-1} = r(\alpha(v)) r(\alpha(w))$ lies in $SO(2)$, hence has the form $\rho(z) = r(z) r(e_1)$ for some $z \in \mathbb{R}^2$ by statement (2) for $n = 2$. Therefore

$$r(v) r(w) = \alpha^{-1} r(z) r(e_1) \alpha = r(\alpha^{-1}(z)) r(\alpha^{-1}(e_1)) = r(x) r(y)$$

for $x = \alpha^{-1}(z) \in \mathbb{R}^{i+1}$ and $y \in \mathbb{R}^i$.

It remains to show that the map $\rho: P^{n-1} \times P^{n-2} \times \cdots \times P^1 \rightarrow SO(n)$ is cellular. This follows from the inclusions $P^i P^i \subset P^i P^{i-1}$ derived above, together with another family of inclusions $P^i P^j \subset P^j P^i$ for $i < j$. To prove the latter we have the formulas

$$\begin{aligned} \rho(v) \rho(w) &= r(v) r(w') \quad \text{where } w' = r(e_1) w, \text{ as earlier} \\ &= r(v) r(w') r(v) r(v) \\ &= r(r(v) w') r(v) \quad \text{from } r(\alpha(v)) = \alpha r(v) \alpha^{-1} \\ &= r(r(v) r(e_1) w) r(v) = r(\rho(v) w) r(v) \\ &= \rho(\rho(v) w) \rho(v') \quad \text{where } v' = r(e_1) v, \text{ hence } v = r(e_1) v' \end{aligned}$$

In particular, taking $v \in \mathbb{R}^{i+1}$ and $w \in \mathbb{R}^{j+1}$ with $i < j$, we have $\rho(v) w \in \mathbb{R}^{j+1}$, and the product $\rho(v) \rho(w) \in P^i P^j$ equals the product $\rho(\rho(v) w) \rho(v') \in P^j P^i$. $\square$

Mod 2 Homology and Cohomology

Each cell of $SO(n)$ is the homeomorphic image of a cell in $P^{n-1} \times P^{n-2} \times \cdots \times P^1$, so the cellular chain map induced by $\rho: P^{n-1} \times P^{n-2} \times \cdots \times P^1 \rightarrow SO(n)$ is surjective. It follows that with $\mathbb{Z}_2$ coefficients the cellular boundary maps for $SO(n)$ are all trivial since this is true in P^i and hence in $P^{n-1} \times P^{n-2} \times \cdots \times P^1$ by Proposition 3B.1. Thus $H_*(SO(n); \mathbb{Z}_2)$ has a $\mathbb{Z}_2$ summand for each cell of $SO(n)$. One can rephrase this

as saying that there are isomorphisms $H_i(SO(n); \mathbb{Z}_2) \approx H_i(S^{n-1} \times S^{n-2} \times \cdots \times S^1; \mathbb{Z}_2)$ for all i since this product of spheres also has cells in one-to-one correspondence with admissible sequences. The full structure of the $\mathbb{Z}_2$ homology and cohomology rings is given by:

Theorem 3D.2. (a) $H^*(SO(n); \mathbb{Z}_2) \approx \bigotimes_{i \text{ odd}} \mathbb{Z}_2[\beta_i]/(\beta_i^{p_i})$ where $|\beta_i| = i$ and p_i is the smallest power of 2 such that $|\beta_i^{p_i}| \geq n$.
(b) The Pontryagin ring $H_*(SO(n); \mathbb{Z}_2)$ is the exterior algebra $\Lambda_{\mathbb{Z}_2}[e^1, \dots, e^{n-1}]$.

Here e^i denotes the cellular homology class of the cell $e^i \subset P^{n-1} \subset SO(n)$, and β_i is the dual class to e^i , represented by the cellular cochain assigning the value 1 to the cell e^i and 0 to all other i -cells.

Proof: As we noted above, ρ induces a surjection on cellular chains. Since the cellular boundary maps with $\mathbb{Z}_2$ coefficients are trivial for both $P^{n-1} \times \cdots \times P^1$ and $SO(n)$, it follows that ρ_* is surjective on $H_*(-; \mathbb{Z}_2)$ and ρ^* is injective on $H^*(-; \mathbb{Z}_2)$. We know that $H^*(P^{n-1} \times \cdots \times P^1; \mathbb{Z}_2)$ is the polynomial ring $\mathbb{Z}_2[\alpha_1, \dots, \alpha_{n-1}]$ truncated by the relations $\alpha_i^{i+1} = 0$. For $\beta_i \in H^i(SO(n); \mathbb{Z}_2)$ the dual class to e^i , we have $\rho^*(\beta_i) = \sum_j \alpha_j^i$, the class assigning 1 to each i -cell in a factor P^j of $P^{n-1} \times \cdots \times P^1$ and 0 to all other i -cells, which are products of lower-dimensional cells and hence map to cells in $SO(n)$ disjoint from e^i .

First we will show that the monomials $\beta_I = \beta_{i_1} \cdots \beta_{i_m}$ corresponding to admissible sequences I are linearly independent in $H^*(SO(n); \mathbb{Z}_2)$, hence are a vector space basis. Since ρ^* is injective, we may identify each β_i with its image $\sum_j \alpha_j^i$ in the truncated polynomial ring $\mathbb{Z}_2[\alpha_1, \dots, \alpha_{n-1}]/(\alpha_1^2, \dots, \alpha_{n-1}^n)$. Suppose we have a linear relation $\sum_I b_I \beta_I = 0$ with $b_I \in \mathbb{Z}_2$ and I ranging over the admissible sequences. Since each β_I is a product of distinct β_i 's, we can write the relation in the form $x\beta_1 + y = 0$ where neither x nor y has β_1 as a factor. Since α_1 occurs only in the term β_1 of $x\beta_1 + y$, where it has exponent 1, we have $x\beta_1 + y = x\alpha_1 + z$ where neither x nor z involves α_1 . The relation $x\alpha_1 + z = 0$ in $\mathbb{Z}_2[\alpha_1, \dots, \alpha_{n-1}]/(\alpha_1^2, \dots, \alpha_{n-1}^n)$ then implies $x = 0$. Thus we may assume the original relation does not involve β_1 . Now we repeat the argument for β_2 . Write the relation in the form $x\beta_2 + y = 0$ where neither x nor y involves β_2 or β_1 . The variable α_2 now occurs only in the term β_2 of $x\beta_2 + y$, where it has exponent 2, so we have $x\beta_2 + y = x\alpha_2^2 + z$ where x and z do not involve α_1 or α_2 . Then $x\alpha_2^2 + z = 0$ implies $x = 0$ and we have a relation involving neither β_1 nor β_2 . Continuing inductively, we eventually deduce that all coefficients b_I in the original relation $\sum_I b_I \beta_I = 0$ must be zero.

Observe now that $\beta_i^2 = \beta_{2i}$ if $2i < n$ and $\beta_i^2 = 0$ if $2i \geq n$, since $(\sum_j \alpha_j^i)^2 = \sum_j \alpha_j^{2i}$. The quotient Q of the algebra $\mathbb{Z}_2[\beta_1, \beta_2, \dots]$ by the relations $\beta_i^2 = \beta_{2i}$ and $\beta_j = 0$ for $j \geq n$ then maps onto $H^*(SO(n); \mathbb{Z}_2)$. This map $Q \rightarrow H^*(SO(n); \mathbb{Z}_2)$ is also injective since the relations defining Q allow every element of Q to be represented as a linear combination of admissible monomials β^I , and the admissible

monomials are linearly independent in $H^*(SO(n); \mathbb{Z}_2)$. The algebra Q can also be described as the tensor product in statement (a) of the theorem since the relations $\beta_i^2 = \beta_{2i}$ allow admissible monomials to be written uniquely as monomials in powers of the β_i 's with i odd, and the relation $\beta_j = 0$ for $j \geq n$ becomes $\beta_{ip_i} = \beta_i^{p_i} = 0$ where $j = ip_i$ with i odd and p_i a power of 2. For a given i , this relation holds iff $ip_i \geq n$, or in other words, iff $|\beta_i^{p_i}| \geq n$. This finishes the proof of (a).

For part (b), note first that the group multiplication $SO(n) \times SO(n) \rightarrow SO(n)$ is cellular in view of the inclusions $P^i P^i \subset P^i P^{i-1}$ and $P^i P^j \subset P^j P^i$ for $i < j$. So we can compute Pontryagin products at the cellular level. We know that there is at least an additive isomorphism $H_*(SO(n); \mathbb{Z}_2) \approx \Lambda_{\mathbb{Z}_2}[e^1, \dots, e^{n-1}]$ since the products $e^I = e^{i_1} \dots e^{i_m}$ with I admissible form a basis for $H_*(SO(n); \mathbb{Z}_2)$. The inclusion $P^i P^i \subset P^i P^{i-1}$ then implies that the Pontryagin product $(e^i)^2$ is 0. It remains only to see the commutativity relation $e^i e^j = e^j e^i$. The inclusion $P^i P^j \subset P^j P^i$ for $i < j$ was obtained from the formula $\rho(v)\rho(w) = \rho(\rho(v)w)\rho(v')$ for $v \in \mathbb{R}^{i+1}$, $w \in \mathbb{R}^{j+1}$, and $v' = r(e_1)v$. The map $f: P^i \times P^j \rightarrow P^j \times P^i$, $f(v, w) = (\rho(v)w, v')$, is a homeomorphism since it is the composition of homeomorphisms $(v, w) \mapsto (v, \rho(v)w) \mapsto (v', \rho(v)w) \mapsto (\rho(v)w, v')$. The first of these maps takes $e^i \times e^j$ homeomorphically onto itself since $\rho(v)(e^j) = e^j$ if $i < j$. Obviously the second map also takes $e^i \times e^j$ homeomorphically onto itself, while the third map simply transposes the two factors. Thus f restricts to a homeomorphism from $e^i \times e^j$ onto $e^j \times e^i$, and therefore $e^i e^j = e^j e^i$ in $H_*(SO(n); \mathbb{Z}_2)$. $\square$

The cup product and Pontryagin product structures in this theorem may seem at first glance to be unrelated, but in fact the relationship is fairly direct. As we saw in the previous section, the dual of a polynomial algebra $\mathbb{Z}_2[x]$ is a divided polynomial algebra $\Gamma_{\mathbb{Z}_2}[\alpha]$, and with $\mathbb{Z}_2$ coefficients the latter is an exterior algebra $\Lambda_{\mathbb{Z}_2}[\alpha_0, \alpha_1, \dots]$ where $|\alpha_i| = 2^i |x|$. If we truncate the polynomial algebra by a relation $x^{2^n} = 0$, then this just eliminates the generators α_i for $i \geq n$. In view of this, if it were the case that the generators β_i for the algebra $H^*(SO(n); \mathbb{Z}_2)$ happened to be primitive, then $H^*(SO(n); \mathbb{Z}_2)$ would be isomorphic as a Hopf algebra to the tensor product of the single-generator Hopf algebras $\mathbb{Z}_2[\beta_i]/(\beta_i^{p_i})$, $i = 1, 3, \dots$, hence the dual algebra $H_*(SO(n); \mathbb{Z}_2)$ would be the tensor product of the corresponding truncated divided polynomial algebras, in other words an exterior algebra as just explained. This is in fact the structure of $H_*(SO(n); \mathbb{Z}_2)$, so since the Pontryagin product in $H_*(SO(n); \mathbb{Z}_2)$ determines the coproduct in $H^*(SO(n); \mathbb{Z}_2)$ uniquely, it follows that the β_i 's must indeed be primitive.

It is not difficult to give a direct argument that each β_i is primitive. The coproduct $\Delta: H^*(SO(n); \mathbb{Z}_2) \rightarrow H^*(SO(n); \mathbb{Z}_2) \otimes H^*(SO(n); \mathbb{Z}_2)$ is induced by the group multiplication $\mu: SO(n) \times SO(n) \rightarrow SO(n)$. We need to show that the value of $\Delta(\beta_i)$ on $e^I \otimes e^J$, which we denote $\langle \Delta(\beta_i), e^I \otimes e^J \rangle$, is the same as the value $\langle \beta_i \otimes 1 + 1 \otimes \beta_i, e^I \otimes e^J \rangle$